Theoretical Foundations for 3D Reconstruction of Sports Games with Limited Visual Data Settings

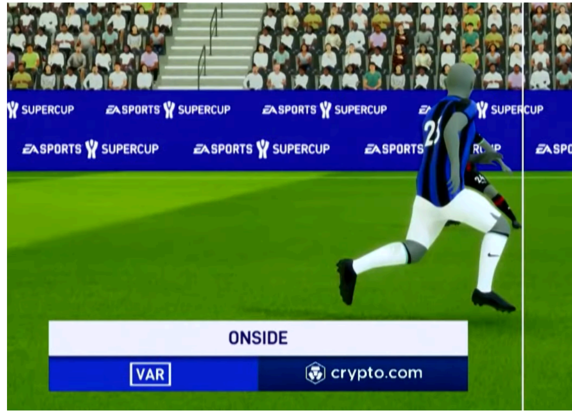
Independent Study Course (NST3901)

Nguyen Hoang The Kiet

Table of Contents

Motivation

Semi-automated offside technology



VAR controversies in the 2024 ASEAN Cup [1], [2]

A suggested pipeline

3-stage pipeline

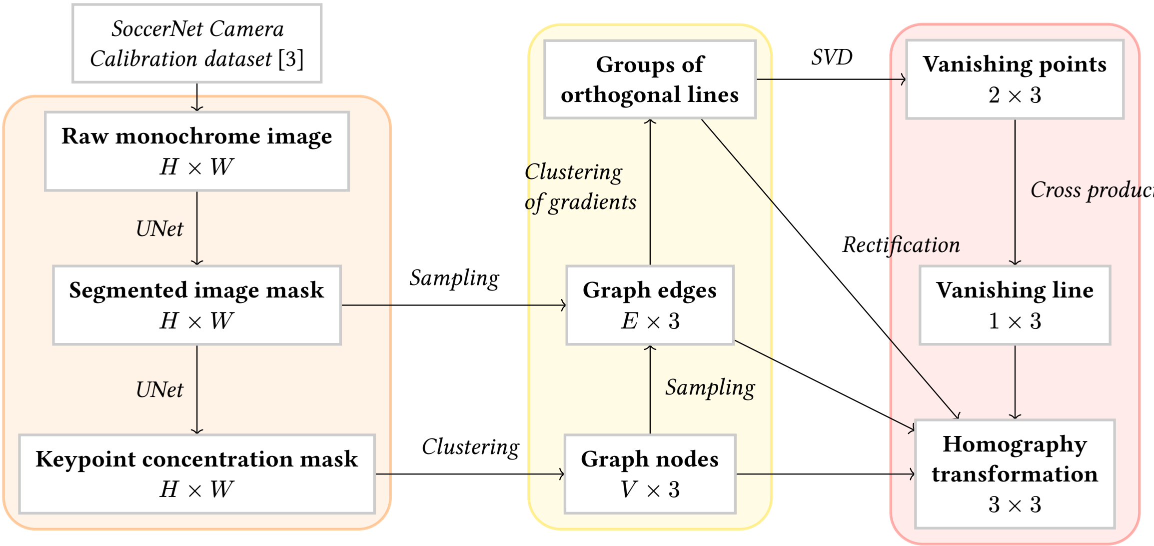
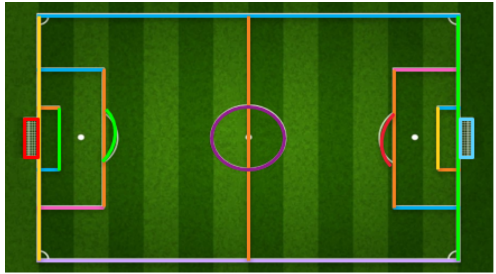


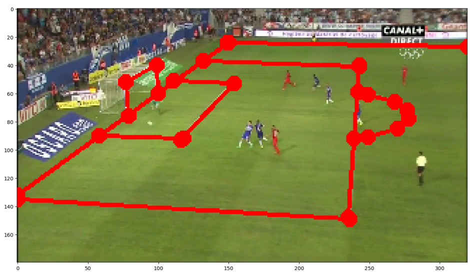
Figure 3: Summary of the suggested automated pipeline for field registration

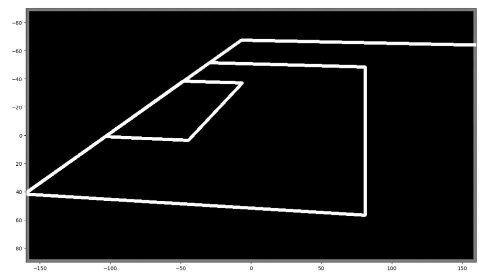
3-stage pipeline

- **Feature masks generation**.
  - Generative image models (U-Net [4])
  - Gray-scale image masks
- **Graph creation**.
  - Keypoints, field lines $\Longrightarrow$ Graph
  - **Orthogonal** field lines
  - Robustness (gamma correction, sampling)
- **Homography reconstruction**.
  - Projective / Multiview geometry [5]
  - Linear algebra

Dataset




CANAL+
DIRECT
0
20
40
60
80
100
120
140
160
0
50
100
150
200
250
300


−80
−60
−40
−20
0
20
40
60
80
−150
−100
−50
0
50
100
150

Dataset (ii)

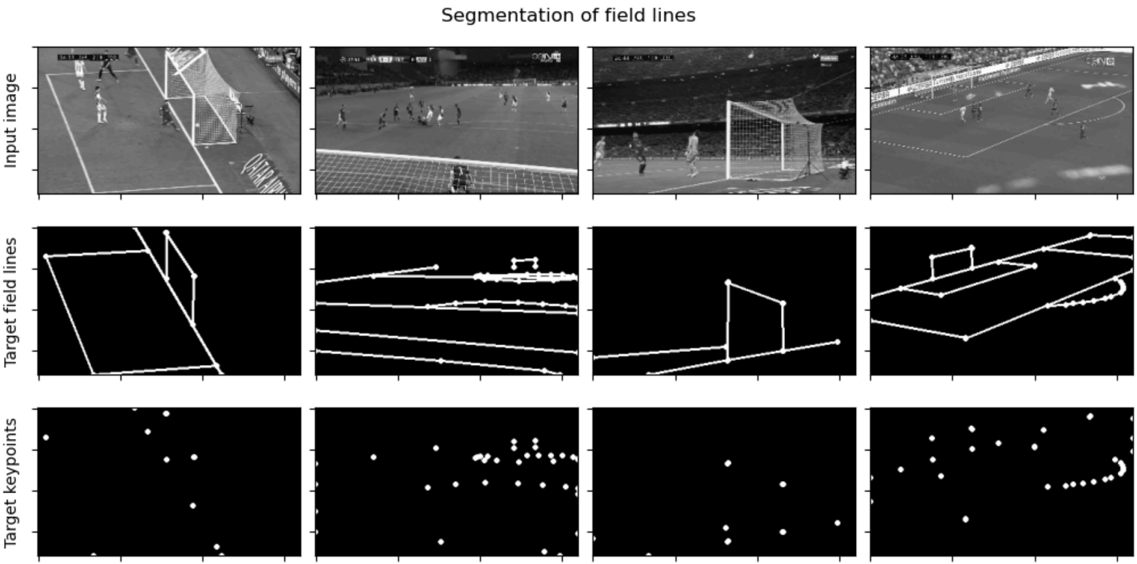
Task / Dataset name	Input	Output / Labels
General `CalibrationDataset`	Raw image $(3 \times H \times W)$	Raw `JSON` dictionary
Edge segmentation `MaskingCalibrationDataset`	Grayscaled image $(H \times W)$	Image mask, highlighted field lines $(H \times W)$

Keypoint concentration
`MaskingCalibrationDataset`
`(keypoint_only=True)`
	Grayscaled image
	$(H \times W)$
		Image mask, highlighted keypoints
		$(H \times W)$

Keypoints `PitchKeypointCalibrationDataset`	Grayscaled image $(H \times W)$	List of all keypoints, reconstructed via homography (27×3)
Homography reconstruction `PitchHomographyCalibrationDataset`	Grayscaled image $(H \times W)$	The homography matrix (3×3)

Stage 1. Image segmentation

Segmentation of field lines and keypoints



UNet [4]: A generative model for feature extraction

- Strong in **localisation** tasks
  - Convolutional layers
  - Detecting patterns based on the relationship between neighbouring pixels
- Skipping neurons
  - Perserves global properties
- Won ISBI 2015 EM segmentation challenge in low-data conditions
  - Ability to learn even with the lack of data

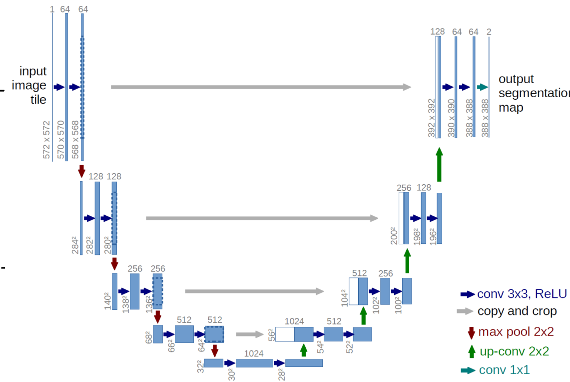


Figure 6: UNet's architecture [4]

Choosing a loss function for segmentation tasks [6]

- Cross Binary Entropy Loss

$$\mathcal{L}_{\mathrm{BCE}}(\vec{\boldsymbol{x}}, \vec{\boldsymbol{y}}) = -\sum_i \log(x_i y_i) = -\log(\vec{\boldsymbol{x}}) \cdot \log(\vec{\boldsymbol{y}})$$

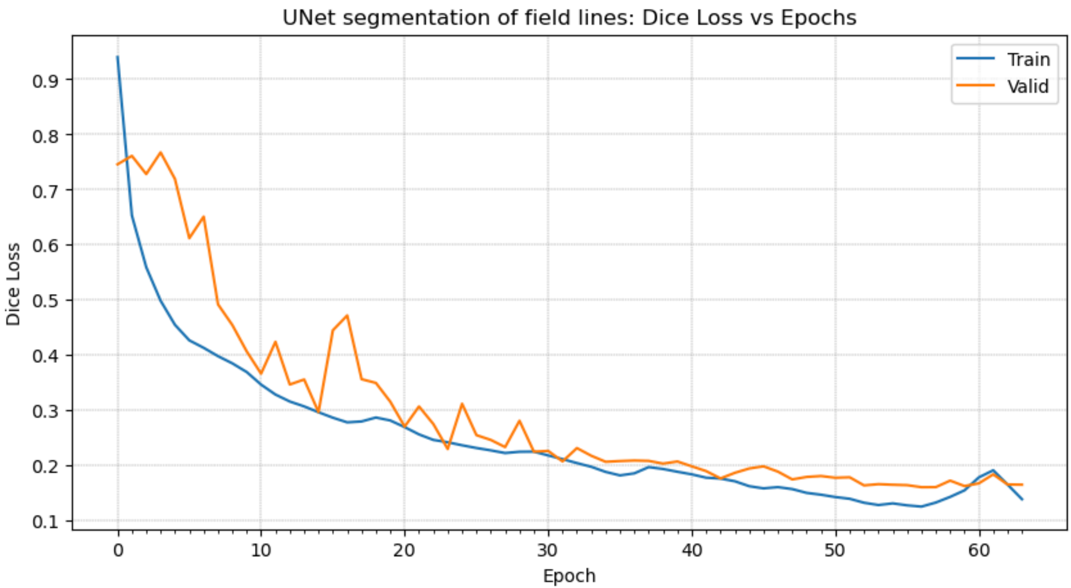
- Intersection-over-Union (IOU) Loss

$$\mathcal{L}_{\mathrm{IOU}}(\vec{\boldsymbol{x}}, \vec{\boldsymbol{y}}) = \frac{|X \cap Y|}{|X \cup Y|} = \frac{\vec{\boldsymbol{x}} \cdot \vec{\boldsymbol{y}}}{1 - \left(\vec{\mathbb{1}} - \vec{\boldsymbol{x}}\right)\left(\vec{\mathbb{1}} - \vec{\boldsymbol{y}}\right)}$$

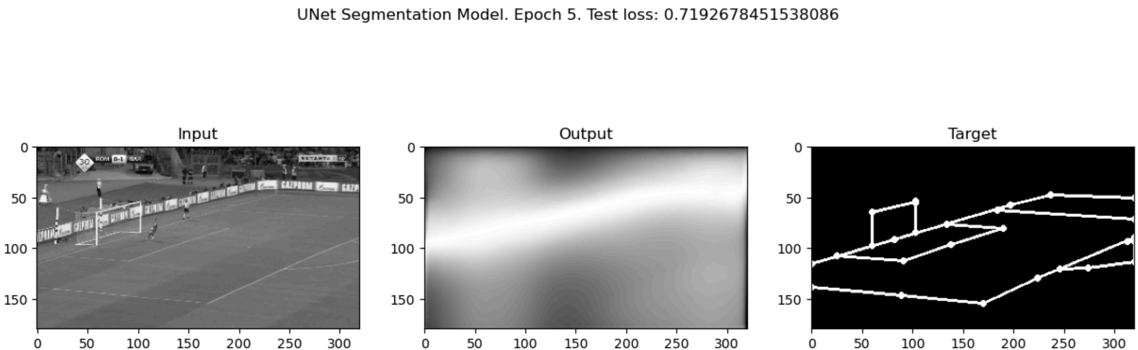
- Dice Loss (for binary classes)

$$\mathcal{L}_{\mathrm{Dice}}(\vec{\boldsymbol{x}}, \vec{\boldsymbol{y}}) = \frac{2|X \cap Y|}{|X| + |Y|} = 2 \cdot \frac{\vec{\boldsymbol{x}} \cdot \vec{\boldsymbol{y}}}{\vec{\boldsymbol{x}} \cdot \vec{\mathbb{1}} + \vec{\boldsymbol{y}} \cdot \vec{\mathbb{1}}}$$

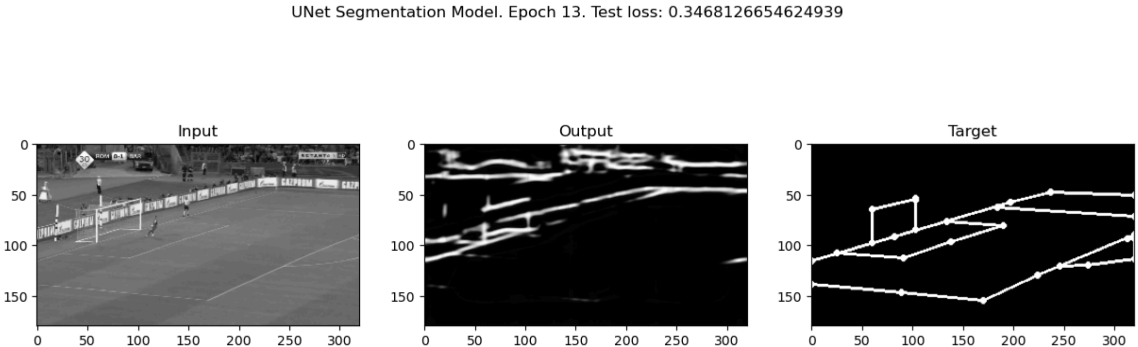
Training metrics (field lines, n_epochs = 64)



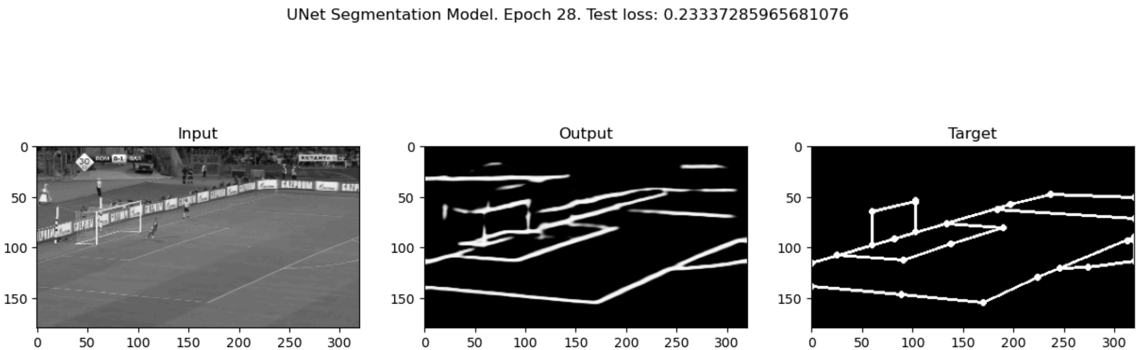
Qualitative results (field lines, `epoch = 05`)



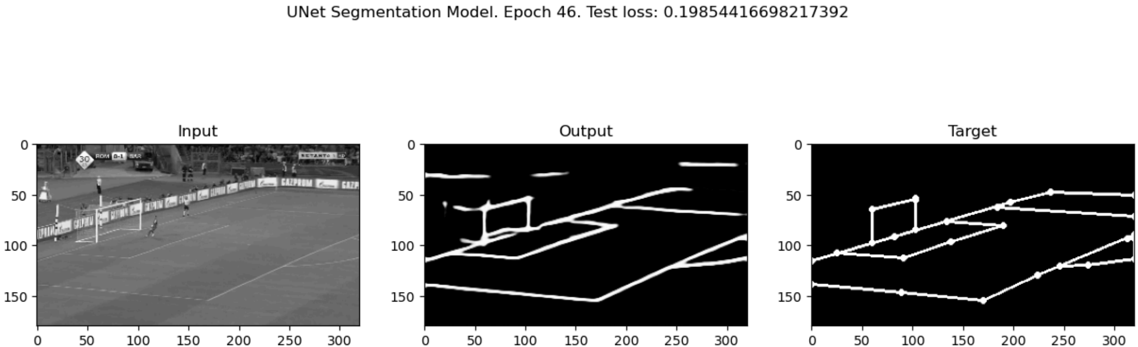
Qualitative results (field lines, epoch = 13)



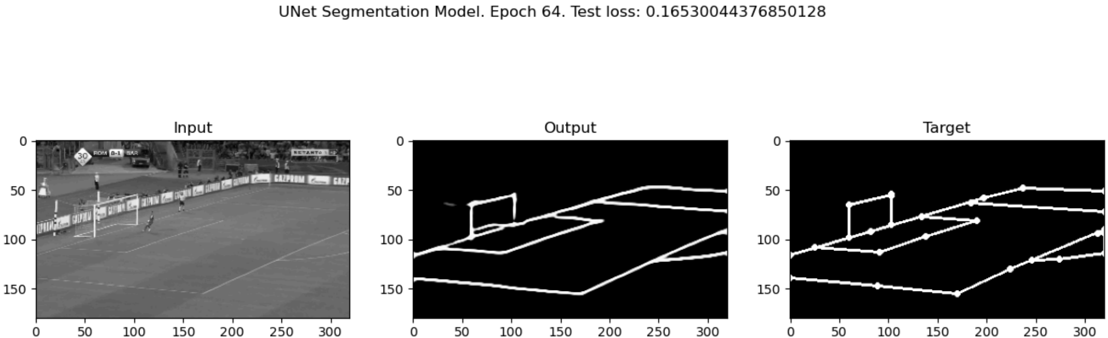
Qualitative results (field lines, epoch = 28)



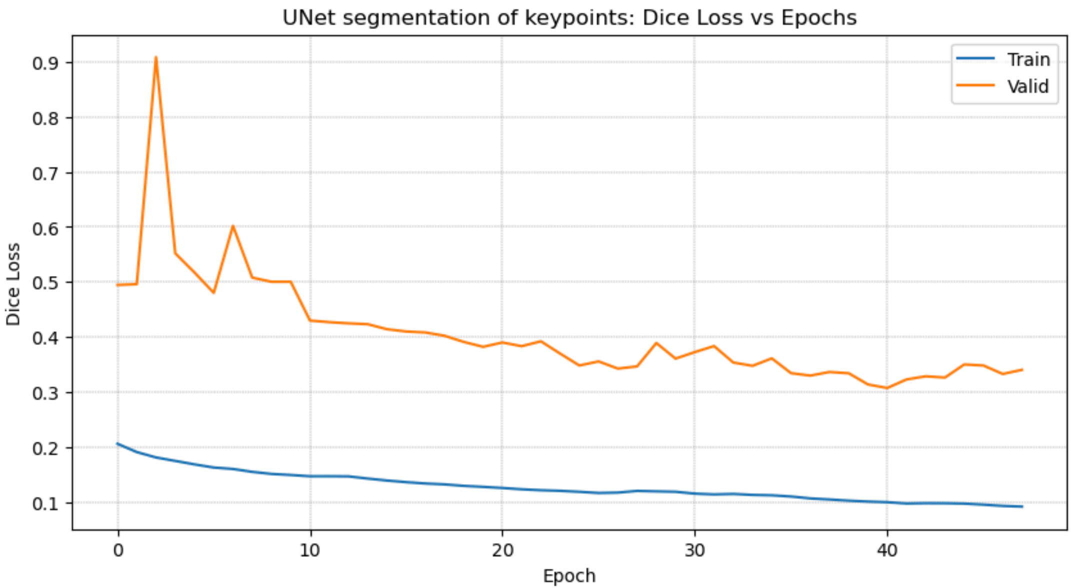
Qualitative results (field lines, epoch = 46)



Qualitative results (field lines, epoch = 64)

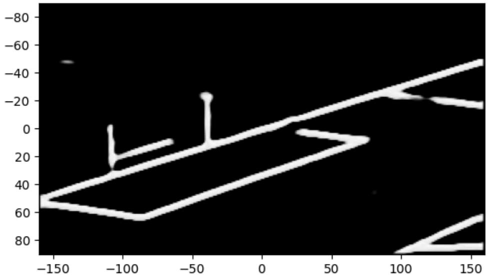


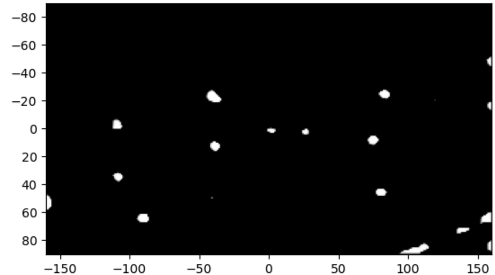
Training metrics (keypoints, n_epochs = 48)



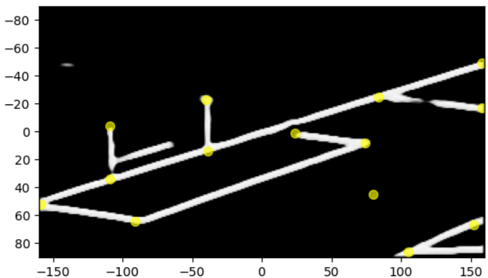
Stage 2. Graph creation

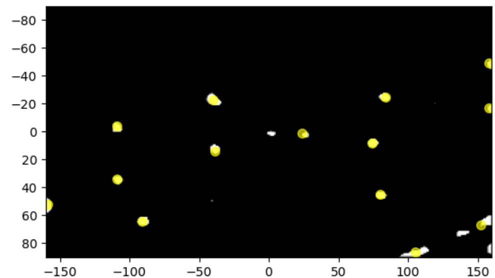
Keypoint detection via pixel clustering



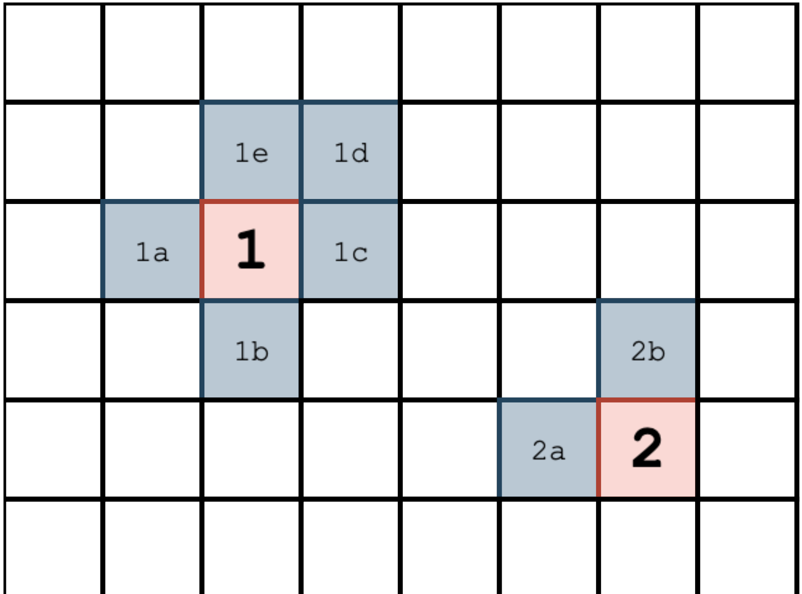


Keypoint detection via pixel clustering

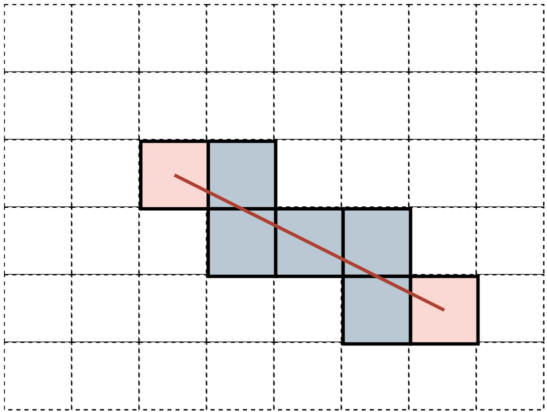


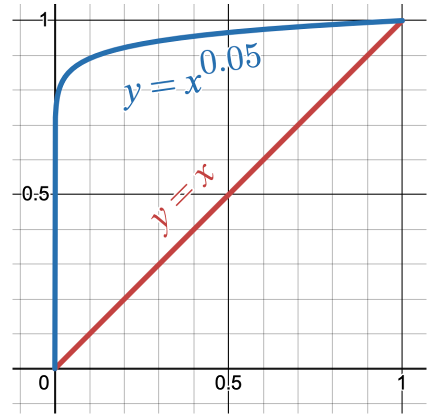


Keypoint detection via pixel clustering



Edge sampling





Gamma correction plot, $\gamma = 0.05$

- Edge mask I_E of size $H \times W$;
- Two keypoints $\vec{\boldsymbol{p}}_1$ and $\vec{\boldsymbol{p}}_2$;

Edge sampling (ii)

- $\gamma \approx 0.01$

Step 1. Let $T := \lceil \|\vec{\boldsymbol{p}}_1 - \vec{\boldsymbol{p}}_2\| \rceil$.

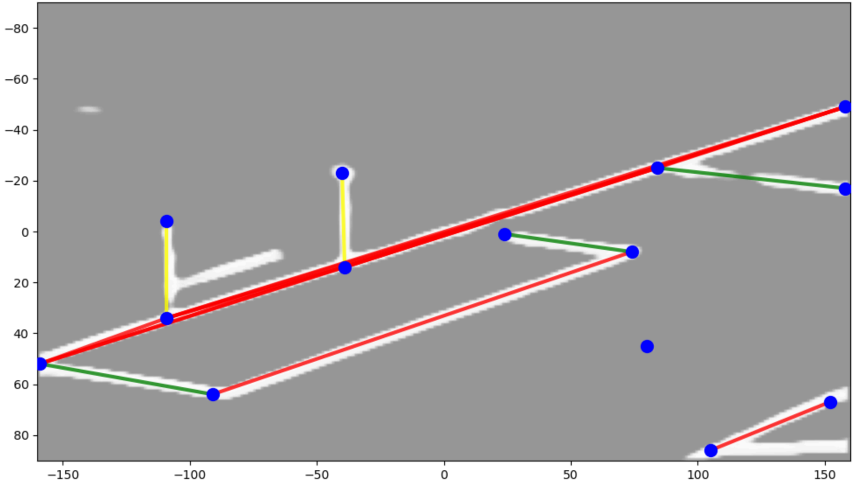
Step 2. Generate T random values of $t^{(i)} \in [0, 1]$.

Step 3. For each $t^{(i)}$, the sampled point is $\vec{\boldsymbol{q}}^{(i)} := \left(1 - t^{(i)}\right)\vec{\boldsymbol{p}}_1 + t^{(i)}\vec{\boldsymbol{p}}_2 + \vec{\boldsymbol{\varepsilon}}$, where $\vec{\boldsymbol{\varepsilon}}$ is a random noise vector, taking range $[-2, 2]$ for each dimension.

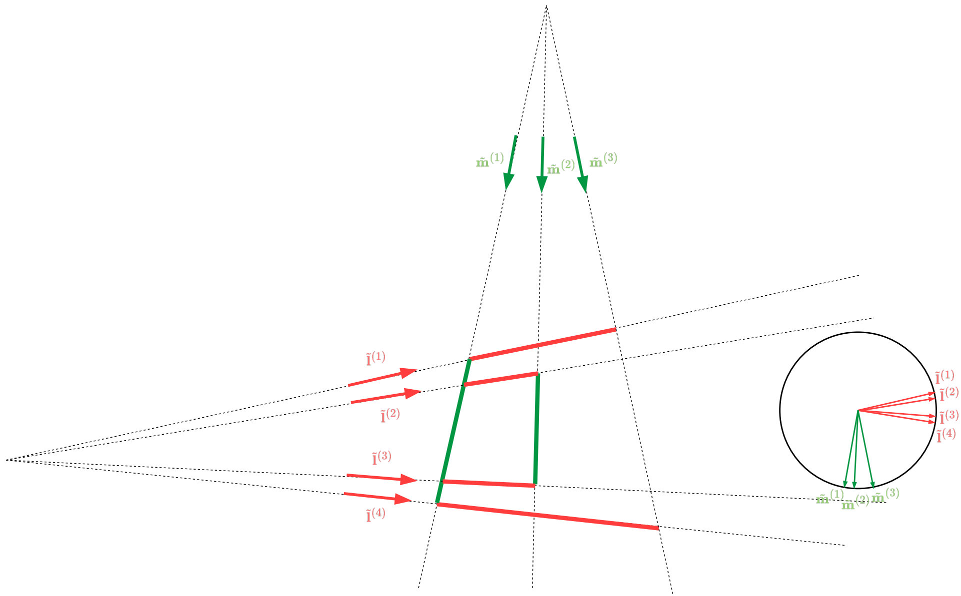
Step 4. Return confidence as the sum of the points' intensities raised to the power of γ:

$$\text{Confidence} := \frac{1}{T} \sum_i I_E\left(q_x^{(i)}, q_y^{(i)}\right)^\gamma$$

Clustering of gradients



Clustering of gradients (ii)



Clustering of gradients (iii)

Grouping lines by the gradient vector.

Input.

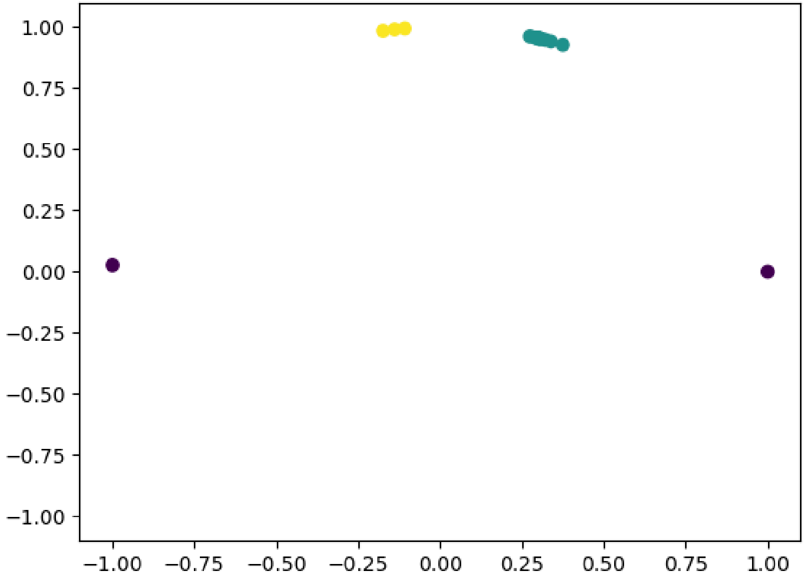
- Set of lines $L = \{\tilde{\boldsymbol{l}}^{(i)}\}$

Output. Two sets of parallel lines L_1 and L_2, where each line in L_1 is orthogonal with a line in L_2 in the original space.

Step 1. For each line $\tilde{\boldsymbol{l}}^{(i)}$, compute the normalised gradient vector $\hat{\boldsymbol{v}}^{(i)} = \frac{1}{\sqrt{{l_1^{(i)}}^2 + {l_2^{(i)}}^2}} \begin{bmatrix} l_1^{(i)} \\ l_2^{(i)} \end{bmatrix}$

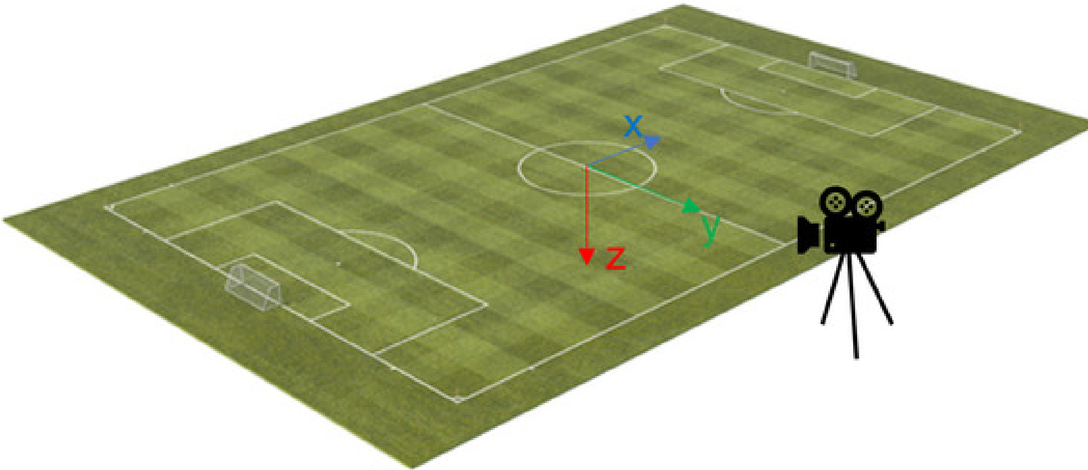
Step 2. Use a clustering algorithm $\mathcal{C}$ to group the lines into two groups.

Clustering of gradients (iv)

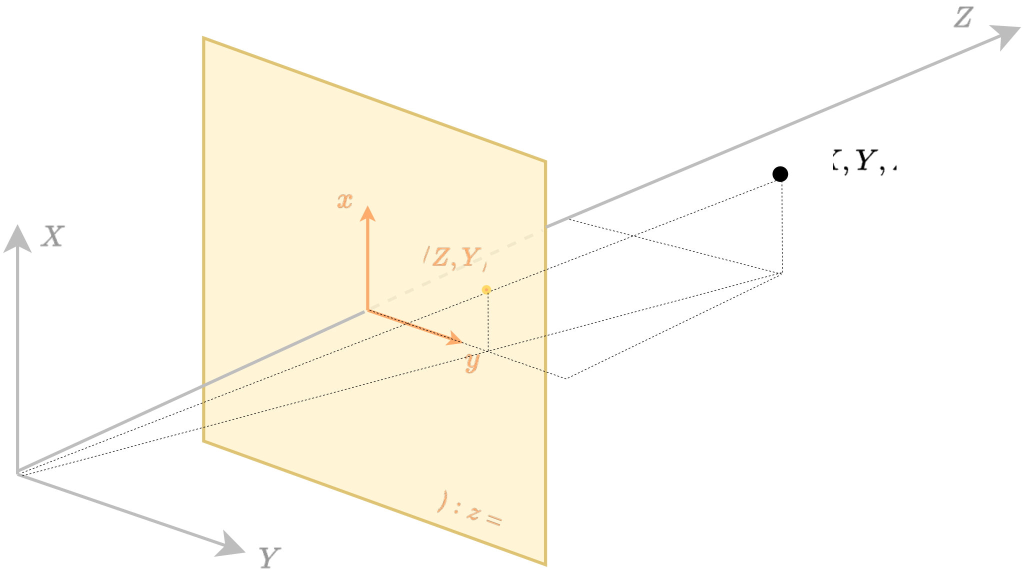


Stage 3. Homography reconstruction

SoccerNet camera model [3]



Projective geometry



Projective geometry (ii)

- Homogenous representation of a point and a line:

$$\tilde{\boldsymbol{p}} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}; \tilde{\boldsymbol{l}} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

- Incidence:

$$(p) \in (\ell) \Leftrightarrow \tilde{\boldsymbol{l}}^T \tilde{\boldsymbol{x}} = 0$$

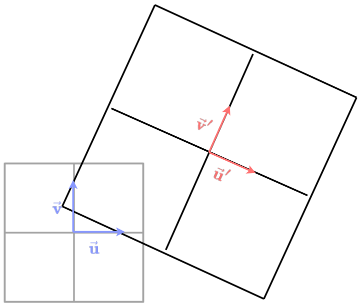
- Line passing two points:

$$\tilde{\boldsymbol{p}} = \tilde{\boldsymbol{l}}_1 \times \tilde{\boldsymbol{l}}_2$$

- Intersection of two lines:

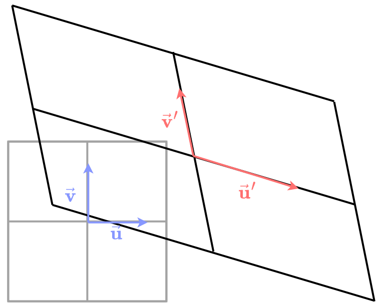
$$\tilde{\boldsymbol{l}} = \tilde{\boldsymbol{x}}_1 \times \tilde{\boldsymbol{x}}_2$$

Transformations in 2D



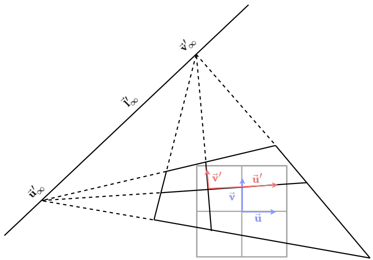
$$\tilde{\boldsymbol{p}}' = \begin{bmatrix} k\cos\theta & -k\sin\theta & v_x \\ k\sin\theta & k\cos\theta & v_y \\ 0 & 0 & 1 \end{bmatrix} \tilde{\boldsymbol{p}}$$

Similarity



$$\tilde{\boldsymbol{p}}' = \begin{bmatrix} a_{11} & a_{12} & v_x \\ a_{21} & a_{22} & v_y \\ 0 & 0 & 1 \end{bmatrix} \tilde{\boldsymbol{p}}$$

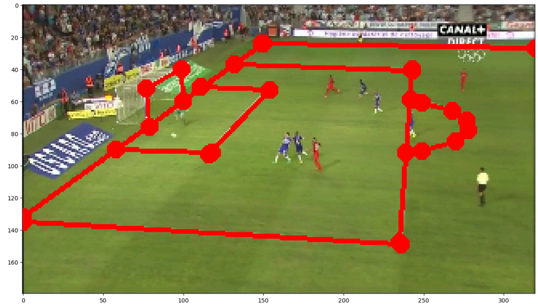
Affinity

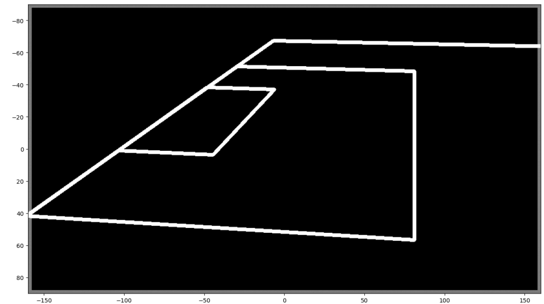


$$\tilde{\boldsymbol{p}}' = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \tilde{\boldsymbol{p}}$$

Homography

Homography reconstruction: Known correspondences

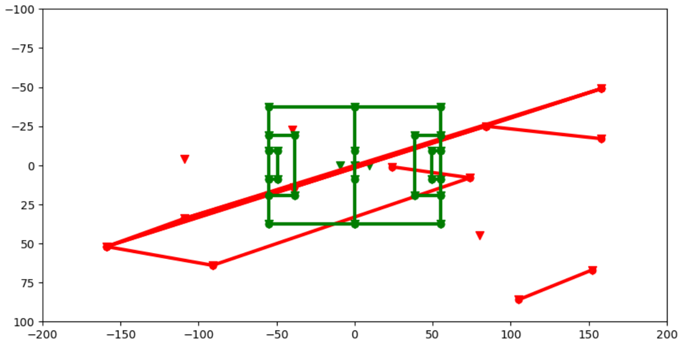




,

- Recover as many unseen keypoints as possible
- Direct Linear Transform (DLT) algorithm (*The 4-point algorithm*)

Homography reconstruction: Unknown correspondences



- Homography rectification
- Affine rectification
- Similarity rectification
- RANSAC

Homography reconstruction: Known correspondences

- Find a homography H given a set of points $P = \left\{\tilde{\boldsymbol{p}}^{(i)}\right\}$ and $Q = \left\{\tilde{\boldsymbol{q}}^{(i)}\right\}$, and that for all i:

$$H : \tilde{\boldsymbol{p}}^{(i)} \mapsto \tilde{\boldsymbol{q}}^{(i)}$$

- Need at least four point to determine such homography
- If more than four points $\longrightarrow$ least squares approach $\longrightarrow$ SVD

Homography reconstruction: Known correspondences (ii)

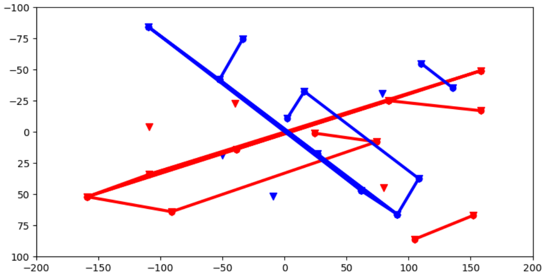
$$H\tilde{\boldsymbol{p}}^{(i)} \times \tilde{\boldsymbol{q}}^{(i)} = \tilde{\boldsymbol{0}}$$

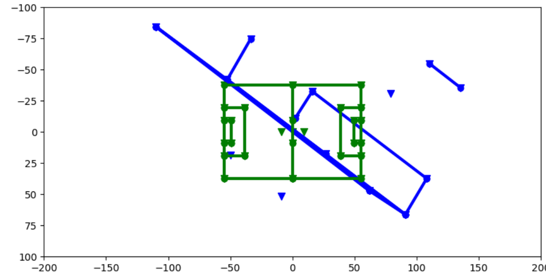
$$\Leftrightarrow \begin{cases} \left(h_{11}p_1^{(i)} + h_{12}p_2^{(i)} + h_{13}p_3^{(i)}\right)q_2^{(i)} - \left(h_{21}p_1^{(i)} + h_{22}p_2^{(i)} + h_{23}p_3^{(i)}\right)q_1^{(i)} = 0 \\ \left(h_{21}p_1^{(i)} + h_{22}p_2^{(i)} + h_{23}p_3^{(i)}\right)q_3^{(i)} - \left(h_{31}p_1^{(i)} + h_{32}p_2^{(i)} + h_{33}p_3^{(i)}\right)q_2^{(i)} = 0 \\ \left(h_{31}p_1^{(i)} + h_{32}p_2^{(i)} + h_{33}p_3^{(i)}\right)q_1^{(i)} - \left(h_{11}p_1^{(i)} + h_{12}p_2^{(i)} + h_{13}p_3^{(i)}\right)q_3^{(i)} = 0 \end{cases}$$

$$\Leftrightarrow \begin{bmatrix} p_1^{(i)}q_2^{(i)} & p_2^{(i)}q_2^{(i)} & p_3^{(i)}q_2^{(i)} & -p_1^{(i)}q_1^{(i)} & -p_2^{(i)}q_1^{(i)} & -p_3^{(i)}q_1^{(i)} & & & \\ & & & p_1^{(i)}q_3^{(i)} & p_2^{(i)}q_3^{(i)} & p_3^{(i)}q_3^{(i)} & -p_1^{(i)}q_2^{(i)} & -p_2^{(i)}q_2^{(i)} & -p_3^{(i)}q_2^{(i)} \\ -p_1^{(i)}q_3^{(i)} & -p_2^{(i)}q_3^{(i)} & -p_3^{(i)}q_3^{(i)} & & & & p_1^{(i)}q_1^{(i)} & p_2^{(i)}q_1^{(i)} & p_3^{(i)}q_1^{(i)} \end{bmatrix} \begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ \vdots \\ h_{31} \\ h_{32} \\ h_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Leftrightarrow A^{(i)}\tilde{\boldsymbol{h}} = \tilde{\boldsymbol{0}}$$

Homography reconstruction: Unknown correspondences





,

Affine and Metric rectification

Similarity rectification

Homography reconstruction: Similarity rectification

- Find a similarity H_S that maps a line segment in the source P to a line segment in the target Q.
- The mapped line segment in P might be the whole segment or a *subsegment* in Q.
- Only need **one** correspondence of segments from P to $Q \Rightarrow$ exhaustive search.
- Which correspondence is the best performing one? How should a correspondence be evaluated?

Homography reconstruction: Similarity rectification

Similarity rectification from a correspondence of two segments.

- *Input*: Four points $\vec{\boldsymbol{p}}^{(1)}, \vec{\boldsymbol{p}}^{(2)}, \vec{\boldsymbol{q}}^{(1)}, \vec{\boldsymbol{q}}^{(2)}$ indicating two segments $\vec{\boldsymbol{p}}^{(1)}, \vec{\boldsymbol{p}}^{(2)}$ and $\vec{\boldsymbol{q}}^{(1)}, \vec{\boldsymbol{q}}^{(2)}$.
- *Output*: The similarity H_S.

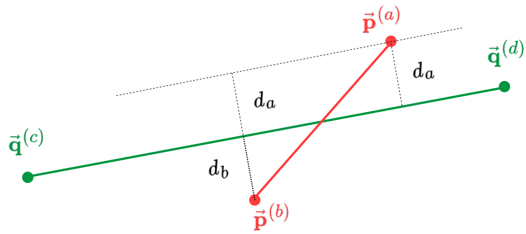
Step 1. Let $\left(p_1^{(i)}, p_2^{(i)}, 1\right) := \tilde{\boldsymbol{p}}^{(i)}$, $\left(q_1^{(i)}, q_2^{(i)}, 1\right) := \tilde{\boldsymbol{q}}^{(i)}$ for $i = 1, 2$.

Step 2. Solve the linear system $H_S \tilde{\boldsymbol{p}}^{(i)} = \tilde{\boldsymbol{q}}^{(i)}$.

$$\begin{cases} ap_1^{(1)} - bp_2^{(1)} + c = q_1^{(1)} \\ ap_2^{(1)} + bp_1^{(1)} + d = q_2^{(1)} \\ ap_1^{(2)} - bp_2^{(2)} + c = q_1^{(2)} \\ ap_2^{(2)} + bp_1^{(2)} + d = q_2^{(2)} \end{cases} \quad \therefore \quad \begin{bmatrix} p_1^{(1)} & -p_2^{(1)} & 1 & 0 \\ p_2^{(1)} & p_1^{(1)} & 0 & 1 \\ p_1^{(2)} & -p_2^{(2)} & 1 & 0 \\ p_2^{(2)} & p_1^{(2)} & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} q_1^{(1)} \\ q_2^{(1)} \\ q_1^{(2)} \\ q_2^{(2)} \end{bmatrix}$$

The similarity can thus be solved using any linear solver such as the SVD.

Homography reconstruction: Similarity rectification



Homography reconstruction: Similarity rectification

Modified RANSAC to find segment correspondence.

Step 1. Let $S_P := \emptyset$ and $S_Q := \emptyset$

Step 2. Select a random segment $\boldsymbol{p}^{(\alpha\beta)} = \left(\vec{\boldsymbol{p}}^{(\alpha)}, \vec{\boldsymbol{p}}^{(\beta)}\right)$ in $\boldsymbol{P}$ and a random segment $\boldsymbol{q}^{(\gamma\delta)} = \left(\vec{\boldsymbol{q}}^{(\gamma)}, \vec{\boldsymbol{q}}^{(\delta)}\right)$.

Step 3. Find a similarity H_S such that the segment $\boldsymbol{p}^{(\alpha\beta)}$ is mapped to $\boldsymbol{q}^{(\gamma\delta)}$.

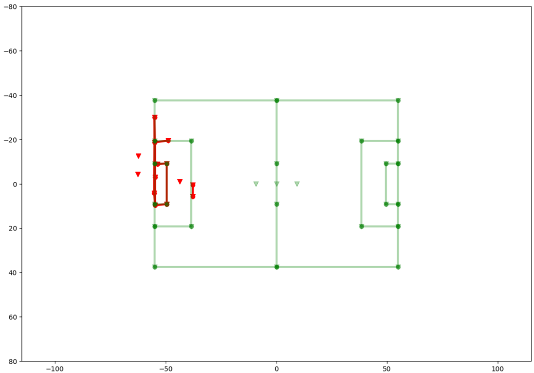
Step 4. For each $\boldsymbol{p}^{(ab)}$ in $\boldsymbol{P}$ and $\boldsymbol{q}^{(cd)}$, calculate the distances between the endpoints and the other segment:

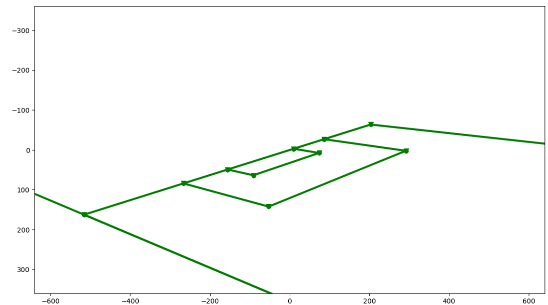
$$d_a := \operatorname{dist}\left(\vec{\boldsymbol{p}}^{(a)}, \boldsymbol{q}^{(cd)}\right); \qquad d_b := \operatorname{dist}\left(\vec{\boldsymbol{p}}^{(b)}, \boldsymbol{q}^{(cd)}\right)$$

- Let $d_{\text{avg}} = \dfrac{1}{2}(d_a + d_b)$. If $d_{\text{avg}} \leq T$, $S_P := S_P \cup \{ab\}$, $S_Q := S_Q \cup \{cd\}$.

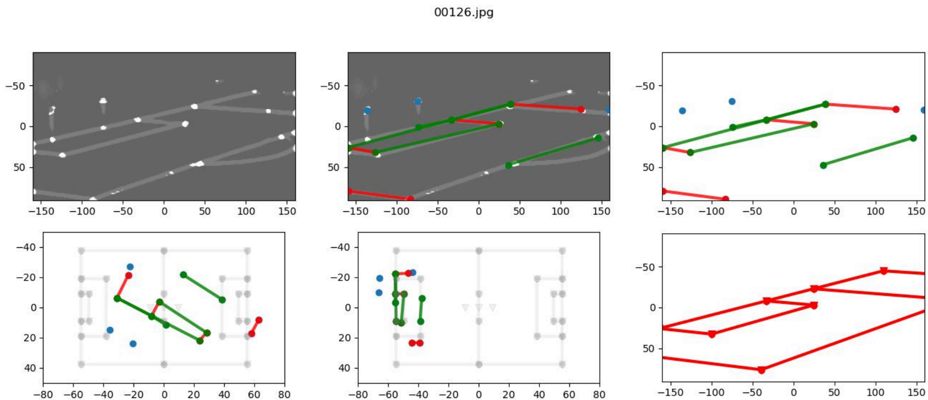
Step 5. Let $\text{Score} = |S_P| + |S_Q|$. If $\text{Score} \geq \text{Score}_{\text{Accept}}$, return H_S. Otherwise, repeat the process.

Results

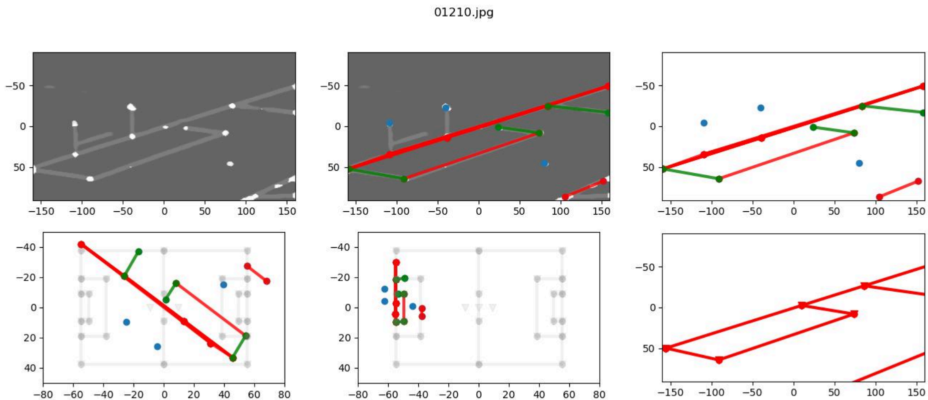

−80
−60
−40
−20
0
20
40
60
80
−100
−50
0
50
100


−300
−200
−100
0
100
200
300
−600
−400
−200
0
200
400
600

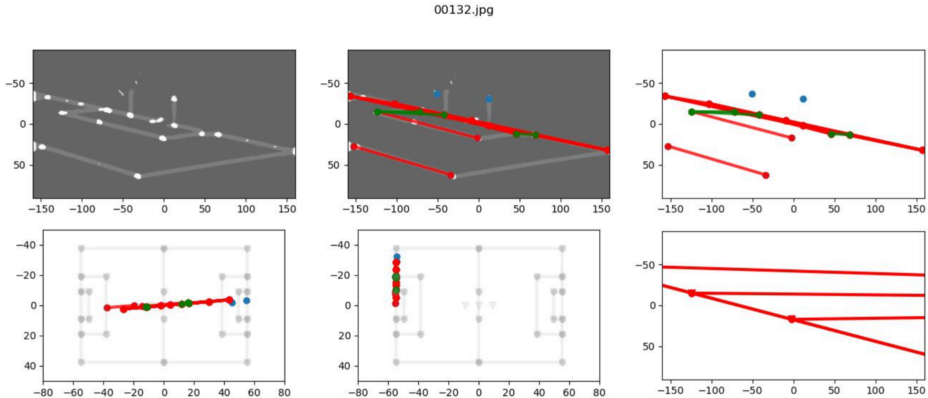
Results (ii)



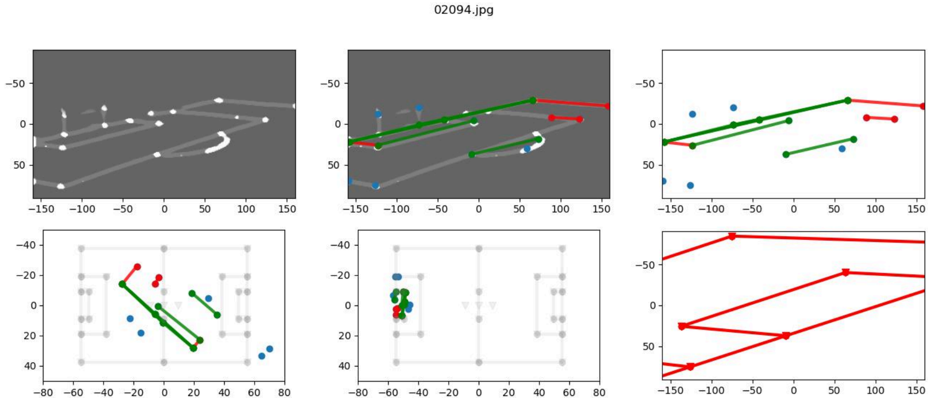
Results (iii)



Results (iv)



Results (v)


02094.jpg
−50
0
50
−150
−100
−50
0
50
100
150
−40
−20
0
20
40
−80
−60
−40
−20
0
20
40
60
80

Discussion

- **Feature masks generation**.
  - Loss function (Regularisation, geometric invariants, etc.)
  - Data augmentation techniques (constrast, brightness, etc.)
- **Graph creation**.
  - More robust algorithms
  - Improving speed (bottleneck at RANSAC)
- **Homography reconstruction**.
  - Numerical stability
  - Different representation of the homography transform (4-point, camera displacement, etc.)

References

[1] ASEAN United FC, "🇹🇭 THAILAND 3-1 PHILIPPINES 🇵🇭 (4-3 AGG.) | #MITSUBISHIELECTRICCUP 2024 SEMI-FINAL LEG 2." Accessed: July 26, 2025. [Online]. Available: https://www.youtube.com/watch?v=aoC8EJ6QHuQ

[2] ASEAN United FC, "🇵🇭 PHILIPPINES 2-1 THAILAND 🇹🇭 | #MITSUBISHIELECTRICCUP 2024 SEMI-FINAL LEG 1." Accessed: July 26, 2025. [Online]. Available: https://www.youtube.com/watch?v=mk0tK4X_KqQ

[3] "SoccerNet - 2025." Accessed: July 26, 2025. [Online]. Available: https://www.soccer-net.org/challenges/2025

[4] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation." Accessed: Nov. 03, 2025. [Online]. Available: http://arxiv.org/abs/1505.04597

[5] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*. Cambridge University Press, 2003.

[6] "Loss Functions in the Era of Semantic Segmentation: A Survey and Outlook." Accessed: Nov. 08, 2025. [Online]. Available: https://arxiv.org/html/2312.05391